



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

PHYSICAL SCIENCES P1

MARKING GUIDELINE

MARKS: 150

This marking guideline consists of 5 pages.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept correct alternative methods. Give follow-through marks (CA) where a learner correctly uses an earlier wrong answer. Penalise once per question for missing or wrong units in final answers.

QUESTION 1

1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
C	A	B	C	C	D	D	B	B	A
✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓

[20]

QUESTION 2

- 2.1 A physical quantity that has both magnitude ✓ and direction ✓. (2)
- 2.2 The change in position of an object ✓✓ (a straight line from start to end, with direction). (2)
- 2.3 Scalar ✓ (1)
- 2.4 $300 + 400 = 700 \text{ m}$ ✓ (1)
- 2.5 $\Delta x^2 = 300^2 + 400^2$ ✓ $\rightarrow \Delta x = \sqrt{250\,000}$ ✓ $= 500 \text{ m}$ ✓ (3)
- 2.6 $\tan \theta = \frac{400}{300}$ ✓ $\rightarrow \theta = 53,13^\circ$ north of east ✓ (2)
- 2.7 $\text{average speed} = \frac{\text{distance}}{\text{time}} = \frac{700}{500}$ ✓✓ $= 1,4 \text{ m} \cdot \text{s}^{-1}$ ✓ (3)
- 2.8 $v_{\text{ave}} = \frac{\Delta x}{\Delta t} = \frac{500}{500}$ ✓ $= 1 \text{ m} \cdot \text{s}^{-1}$ ✓ at $53,13^\circ$ north of east ✓ (3)

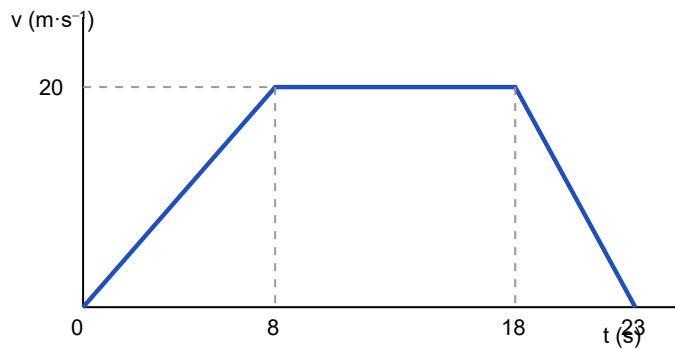
[17]

QUESTION 3

- 3.1 The rate of change of velocity ✓✓. (2)
- 3.2 $v_f = v_i + a\Delta t = 0 + (2,5)(8)$ ✓✓ $= 20 \text{ m} \cdot \text{s}^{-1}$ (in the direction of motion) ✓ (3)
- 3.3 $\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2 = 0 + \frac{1}{2}(2,5)(8)^2$ ✓✓ $= 80 \text{ m}$ ✓ (3)
- 3.4 $a = \frac{v_f - v_i}{\Delta t} = \frac{0 - 20}{5}$ ✓✓ $= -4 \text{ m} \cdot \text{s}^{-2}$ ($4 \text{ m} \cdot \text{s}^{-2}$ opposite to the motion) ✓ (3)

3.5

(4)



Marks: straight line from (0 ; 0) up to 20 m·s⁻¹ at 8 s ✓; horizontal line from 8 s to 18 s ✓; straight line down to 0 at 23 s ✓; values and labelled axes ✓

3.6

Displacement = area under graph ✓

(4)

$$= \frac{1}{2}(8)(20) + (10)(20) + \frac{1}{2}(5)(20) \checkmark = 80 + 200 + 50 \checkmark = 330 \text{ m } \checkmark$$

3.7

$$v_{\text{ave}} = \frac{\Delta x}{\Delta t} = \frac{330}{23} \checkmark \checkmark = 14,35 \text{ m} \cdot \text{s}^{-1} \text{ in the direction of motion } \checkmark$$

(3)

[22]**QUESTION 4**

4.1

In the absence of air resistance (or any external forces), the total mechanical energy of an object remains constant (is conserved) ✓✓.

(2)

4.2

$$E_p = mgh = (0,5)(9,8)(20) \checkmark \checkmark = 98 \text{ J } \checkmark$$

(3)

4.3

 $E_M = 98 \text{ J}$ at all heights ✓

(4)

$$E_p \text{ at } 5 \text{ m} = (0,5)(9,8)(5) = 24,5 \text{ J } \checkmark$$

$$E_K = 98 - 24,5 \checkmark = 73,5 \text{ J } \checkmark$$

4.4

 $E_K = E_M = 98 \text{ J}$ at the ground ✓

(4)

$$\frac{1}{2}(0,5)v^2 = 98 \checkmark \rightarrow v^2 = 392 \checkmark \rightarrow v = 19,80 \text{ m} \cdot \text{s}^{-1} \checkmark$$

4.5

$$E_K = \frac{1}{2}mv^2 = \frac{1}{2}(1\,200)(20)^2 \checkmark \checkmark = 240\,000 \text{ J } (2,4 \times 10^5 \text{ J}) \checkmark$$

(3)

4.6

It becomes 4 times bigger ✓✓ ($E_K \propto v^2$, and $2^2 = 4$).

(2)

[18]**QUESTION 5**

5.1

A wave in which the particles of the medium vibrate perpendicular ✓ to the direction in which the wave moves ✓.

(2)

5.2

$$v = f\lambda = (50)(0,4) \checkmark \checkmark = 20 \text{ m} \cdot \text{s}^{-1} \checkmark$$

(3)

5.3

$$T = \frac{1}{f} = \frac{1}{50} \checkmark \checkmark = 0,02 \text{ s } \checkmark$$

(3)

5.4

When two pulses meet, the resultant displacement is equal to the algebraic sum of the displacements of the individual pulses ✓✓. Amplitude = 3 + 2 = 5 cm (constructive interference) ✓

(3)

- 5.5 Distance travelled = $2 \times 170 = 340 \text{ m}$ ✓ (3)
- $$v = \frac{\text{distance}}{\text{time}} = \frac{340}{1,0} \text{ ✓} = 340 \text{ m} \cdot \text{s}^{-1} \text{ ✓}$$
- 5.6 Longitudinal ✓ (1)
- 5.7 Frequency ✓ (1)
- 5.8 No ✓. Humans can only hear frequencies from about 20 Hz to 20 kHz, and 25 kHz is above this range (ultrasound) ✓. (2)

[18]

QUESTION 6

- 6.1 Radio waves, microwaves, visible light, X-rays ✓✓ (2)
- 6.2 A packet (quantum) of energy ✓ of electromagnetic radiation ✓. (2)
- 6.3 $E = hf = (6,63 \times 10^{-34})(6 \times 10^{14}) \text{ ✓✓} = 3,98 \times 10^{-19} \text{ J ✓}$ (3)
- 6.4 $\lambda = \frac{c}{f} = \frac{3,0 \times 10^8}{6 \times 10^{14}} \text{ ✓✓} = 5 \times 10^{-7} \text{ m ✓}$ (3)
- 6.5 Any ONE: sunburn, skin cancer, damage to the eyes (cataracts), premature ageing of the skin ✓ (1)

[11]

QUESTION 7

- 7.1 The net charge of an isolated system remains constant ✓✓ (charge cannot be created or destroyed). (2)
- 7.2 $Q = \frac{(+8) + (-4)}{2} \text{ ✓} = +2 \text{ nC on each sphere ✓}$ (2)
- 7.3 $\Delta Q \text{ on B} = (+2) - (-4) = 6 \text{ nC ✓}$ (4)
- $$n = \frac{\Delta Q}{q_e} = \frac{6 \times 10^{-9}}{1,6 \times 10^{-19}} \text{ ✓} = 3,75 \times 10^{10} \text{ electrons ✓}$$
- Electrons moved from sphere B to sphere A ✓
- 7.4 The balloon polarises the wall: electrons in the wall are repelled, so the surface near the balloon becomes positive ✓. Opposite charges attract, so the balloon sticks ✓. (2)
- 7.5 Charge builds up on the truck from friction ✓. Earthing lets the charge flow safely to the ground, so no spark can ignite the fuel ✓. (2)

[12]

QUESTION 8

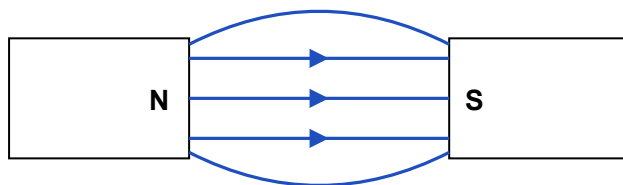
- 8.1 The rate of flow of charge ✓✓. (2)
- 8.2 $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} \text{ ✓} = \frac{1}{2} \text{ ✓} \rightarrow R_p = 2 \text{ } \Omega \text{ ✓}$ (3)
- 8.3 $R_{\text{total}} = 4 + 2 = 6 \text{ } \Omega \text{ ✓}$ (1)

- 8.4 $I = \frac{V}{R} = \frac{12}{6} \checkmark\checkmark = 2 \text{ A } \checkmark$ (3)
- 8.5 $V = IR = (2)(4) \checkmark\checkmark = 8 \text{ V } \checkmark$ (3)
- 8.6 $V_{\text{parallel}} = 12 - 8 = 4 \text{ V } \checkmark; I_3 = \frac{V}{R} = \frac{4}{3} \checkmark = 1,33 \text{ A } \checkmark$ (3)
- 8.7 $Q = I\Delta t = (2)(2 \times 60) \checkmark\checkmark = 240 \text{ C } \checkmark$ (3)
- 8.8 DECREASE $\checkmark$. The total resistance increases ($4 + 3 = 7 \Omega$) $\checkmark$, so for the same emf the current is smaller ($I = V/R = 12/7 = 1,71 \text{ A}$) $\checkmark$. (3)
- 8.9 In parallel across $R_1 \checkmark$ (1)

[22]

QUESTION 9

- 9.1 (3)



Marks: field lines from N to S between the poles $\checkmark$; lines do not cross and are closest where the field is strongest (straight and evenly spaced in the middle) $\checkmark$; arrows pointing from N to S $\checkmark$

- 9.2 The Earth's geographic North Pole is close to a magnetic SOUTH pole $\checkmark$. Opposite poles attract, so the north pole of the compass needle points there $\checkmark$. (2)
- 9.3 Any ONE: place compasses around the wire / sprinkle iron filings on a card around the wire (Oersted's experiment) $\checkmark$ (1)
- 9.4 Point the thumb of the right hand in the direction of the (conventional) current $\checkmark$; the curled fingers show the direction of the magnetic field $\checkmark$. (2)
- 9.5 Any TWO: compass, electric motors, generators, loudspeakers, MRI scanners, fridge magnets, magnetic strips on bank cards, scrapyard electromagnets $\checkmark\checkmark$ (2)

[10]

TOTAL: 150

END OF MARKING GUIDELINE